

RAGHAVENDRA PRASAD VEMURI

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EXECUTIVE SUMMARY

- 13+ years of IT experience with 8+ years specializing in Cloud Architecture, Infrastructure Modernization, and Enterprise DevOps Transformation across Banking and FinTech domains.
- Proven expertise in architecting secure, scalable, and high-availability cloud-native systems on AWS.
- Extensive experience designing enterprise cloud migration strategies including re-platforming, containerization, and microservices modernization.
- Architected distributed systems supporting millions of financial transactions daily.
- Strong background in Infrastructure as Code (Terraform, CloudFormation), CI/CD pipelines, and Kubernetes-based orchestration.
- Designed multi-account AWS environments with centralized security governance and compliance controls.
- Led FinOps initiatives reducing cloud spend by over 28%.
- Deep expertise in VPC architecture, IAM governance, disaster recovery planning, and zero-downtime deployments.

- Experienced in mentoring cross-functional engineering teams and establishing cloud standards across enterprise platforms.
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CORE COMPETENCIES

Enterprise Cloud Architecture
AWS Infrastructure Strategy
Microservices & Containerization
DevOps & CI/CD Architecture
Infrastructure as Code (IaC)
High Availability & Disaster Recovery
Cloud Security & Compliance
Cost Optimization (FinOps)
API & Event-Driven Architecture
Observability & Monitoring Strategy

TECHNICAL SKILLS MATRIX

Cloud Platforms : AWS (EC2, S3, Lambda, RDS, DynamoDB, ECS, EKS, IAM, VPC, CloudFront, Route53, CloudWatch)

Infrastructure as Code : Terraform, AWS CloudFormation

Containers & Orchestration : Docker, Kubernetes (EKS)

CI/CD Tools : Jenkins, GitHub Actions, GitLab CI

Programming & Scripting : Python, Java, Bash, SQL

Databases : PostgreSQL, SQL Server, Oracle, DynamoDB, MongoDB

Monitoring & Logging : Prometheus, Grafana, ELK Stack, CloudWatch

Security & Compliance : IAM Policies, SCP, SSL/TLS, Encryption at Rest & Transit, OAuth2

Networking & Connectivity : VPC Peering, Transit Gateway, Direct Connect, VPN, NACLs, Security Groups, PrivateLink

Serverless & Event-Driven Architecture : API Gateway, SNS, SQS, EventBridge, Step Functions

DevSecOps & Compliance Automation : AWS Config, GuardDuty, Security Hub, SAST/DAST Integration, Secrets Manager

High Availability & Disaster Recovery : Multi-AZ Deployments, Cross-Region Replication, Backup & Restore Strategies, RTO/RPO Planning

FinOps & Cost Optimization : AWS Cost Explorer, Reserved Instances, Spot Instances, Rightsizing, Intelligent Tiering

Microservices & API Architecture : RESTful APIs, API Gateway Design, Service Mesh (Istio – Exposure), Contract-First API Design

Observability & Reliability Engineering : Distributed Tracing, Log Aggregation, Alerting Strategy, SLA/SLO Definition, Chaos Testing (Exposure)

CLOUD ARCHITECTURE EXPERTISE

AWS ENTERPRISE ARCHITECTURE DESIGN

- Designed multi-tier VPC architecture with secure subnet segmentation.
- Implemented cross-account access controls and centralized IAM governance.
- Designed HA systems across multiple availability zones.
- Architected serverless solutions using Lambda & API Gateway.
- Designed centralized logging and observability strategy.
- Designed AWS Landing Zone architecture aligned with enterprise governance models.
- Implemented AWS Control Tower for centralized account provisioning and compliance enforcement.
- Architected multi-region active-active and active-passive failover strategies.
- Designed network segmentation strategy using Transit Gateway and VPC peering.
- Implemented PrivateLink architecture for secure service-to-service

communication.

- Designed scalable event-driven systems using SNS, SQS, and EventBridge.
 - Established centralized identity federation using SSO and IAM role assumptions.
 - Architected secure API architecture with WAF and Shield for DDoS protection.
 - Designed zero-trust network model using strict security group and NACL controls.
 - Implemented infrastructure resilience testing including chaos engineering simulations.
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DEVOPS & AUTOMATION STRATEGY

- Automated infrastructure provisioning using Terraform.
 - Built fully automated CI/CD pipelines for containerized applications.
 - Integrated security scanning and compliance validation in pipelines.
 - Implemented blue-green and rolling deployment strategies.
 - Designed GitOps-based deployment workflows using branch-based infrastructure control.
 - Integrated automated container image scanning using Trivy and ECR scanning.
 - Implemented Terraform modules for reusable infrastructure components.
 - Automated multi-environment provisioning (DEV, QA, UAT, PROD) with configuration isolation.
 - Implemented Canary deployments to reduce release risk.
 - Designed CI/CD rollback mechanisms for zero-downtime recovery.
 - Integrated SonarQube and SAST tools for code quality enforcement.
 - Implemented automated infrastructure compliance validation using AWS Config rules.
 - Designed centralized secrets management using AWS Secrets Manager.
 - Automated database schema migrations as part of deployment workflows.
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ENTERPRISE CLOUD TRANSFORMATION

CLOUD MIGRATION PROGRAM

- Led migration of 35+ on-prem applications to AWS.
 - Designed re-architecture for monolithic systems into microservices.
 - Implemented RDS migration from legacy Oracle databases.
 - Designed disaster recovery with cross-region replication.
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FINOPS & COST OPTIMIZATION

- Reduced EC2 cost through rightsizing initiatives.
 - Implemented auto-scaling policies for dynamic workload handling.
 - Designed reserved instance planning strategy.
 - Reduced overall cloud spend by 28%.
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PROFESSIONAL EXPERIENCE

Goldman Sachs – Senior Cloud Architect

Apr 2020 – Present | Dallas, TX

Project Overview:

Modernization of enterprise trading platform to cloud-native AWS infrastructure.

Key Responsibilities:

Cloud Architecture

- Designed multi-account AWS landing zone.
- Implemented secure VPC peering and network segmentation.
- Designed Kubernetes (EKS) clusters for container workloads.

Microservices & Containers

- Containerized 25+ services using Docker.
- Implemented auto-scaling and high availability policies.

Security & Compliance

- Designed IAM policies following least privilege principles.
- Implemented encryption using AWS KMS.

DevOps & CI/CD

- Integrated Terraform into automated deployment pipelines.
- Reduced deployment time by 45%.

Monitoring & Reliability

- Implemented centralized monitoring with Prometheus & Grafana.
- Achieved 99.99% uptime SLA.

Environment: AWS, Terraform, Kubernetes, Docker, PostgreSQL, Jenkins

JPMorgan Chase – Cloud Solutions Engineer

Jan 2016 – Mar 2020 | Chicago, IL

- Designed AWS infrastructure for customer-facing banking applications.
- Migrated legacy systems to containerized microservices.
- Implemented API Gateway-based secure integrations.
- Automated infrastructure provisioning using CloudFormation.
- Designed Redis caching architecture to improve performance by 35%.
- Designed multi-tier VPC architecture with public/private subnets and NAT gateways for secure internet access.
- Implemented auto-scaling groups and load balancers to handle peak banking transaction traffic.
- Configured AWS IAM roles and policies following least-privilege security principles.
- Implemented RDS multi-AZ deployments ensuring high availability and failover

capability.

- Designed centralized logging solution using CloudWatch Logs and CloudTrail for compliance tracking.
- Implemented encryption for data at rest using AWS KMS and SSL/TLS for data in transit.
- Designed blue-green deployment strategy to minimize downtime during application releases.
- Optimized EC2 instance sizing and storage allocation reducing operational cost by 18%.
- Collaborated with DevOps teams to integrate infrastructure provisioning into CI/CD workflows.
- Conducted disaster recovery testing and validated backup restoration processes across environments.

Environment: AWS, EC2, S3, RDS, Redis, IAM, ELB, Auto Scaling, CloudFormation, Jenkins

Experian – Cloud Infrastructure Engineer

Jun 2012 – Dec 2015 | Atlanta, GA

- Supported hybrid cloud integration strategy.
- Designed secure VPC architecture and security groups.
- Implemented S3 backup and disaster recovery solutions.
- Built CloudWatch dashboards for infrastructure monitoring.
- Assisted in cloud migration planning.
- Implemented EC2 provisioning and AMI management for standardized infrastructure builds.
- Configured security groups and network ACLs to enforce layered security controls.
- Automated infrastructure setup using AWS CLI and scripting tools.
- Designed S3 lifecycle policies for archival and cost optimization.
- Implemented CloudWatch alarms and SNS notifications for proactive monitoring.
- Assisted in migration of on-prem SQL Server databases to AWS RDS.
- Configured Elastic Load Balancers for high availability of web applications.
- Implemented backup strategies using AWS snapshots and cross-region replication.

- Supported infrastructure patch management and OS hardening initiatives.
- Performed root cause analysis for production incidents and implemented preventive measures.

Environment: AWS EC2, S3, RDS, SQL Server, CloudWatch, IAM, Jenkins

MAJOR CLOUD PROGRAMS

Enterprise AWS Modernization Initiative

- Migrated 35+ applications to AWS.
 - Improved system scalability and resilience.
 - Reduced infrastructure provisioning time by 50%.
 - Conducted application dependency mapping and migration readiness assessments.
 - Designed phased migration roadmap minimizing production disruption.
 - Implemented database replication strategy using AWS DMS.
 - Migrated legacy file storage systems to S3 with lifecycle management.
 - Implemented containerization strategy to modernize legacy Java applications.
 - Designed hybrid connectivity using Direct Connect and VPN tunnels.
 - Established cloud governance model including tagging strategy and compliance standards.
 - Conducted performance benchmarking post-migration to validate improvements.
 - Designed API-first modernization strategy for legacy systems.
 - Implemented backup and archival strategy aligned with regulatory requirements.
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Kubernetes Containerization Program

- Containerized legacy services.
- Implemented auto-scaling policies.
- Reduced environment inconsistencies by 70%.

- Implemented AWS Cost Explorer dashboards for granular cost visibility.
 - Designed cost allocation tagging model for business unit accountability.
 - Automated shutdown of idle non-production environments.
 - Implemented spot instance strategies for batch workloads.
 - Optimized S3 storage classes using Intelligent Tiering policies.
 - Conducted monthly cost review governance meetings.
 - Implemented CloudWatch-based usage anomaly detection alerts.
 - Designed container resource allocation tuning to prevent over-provisioning.
 - Optimized RDS instance sizing reducing database costs by 22%.
 - Implemented predictive scaling strategies based on historical workload patterns.
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MEASURABLE IMPACT

- ✓ Reduced cloud infrastructure cost by 28%
 - ✓ Achieved 99.99% availability SLA
 - ✓ Reduced deployment cycle time by 45%
 - ✓ Migrated 35+ enterprise applications
 - ✓ Improved DR recovery time by 40%
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EDUCATION

Master of Science – Cloud Computing
University of Texas at Dallas (2015 – 2017)

Bachelor of Engineering – Computer Science
Osmania University (2008 – 2012)

CERTIFICATIONS

AWS Certified Solutions Architect – Professional
AWS Certified DevOps Engineer
Certified Kubernetes Administrator (CKA)
HashiCorp Terraform Associate